

# Nonparametric Multivariate Location Tests

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3. Paired Samples Test
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Content adapted from:

- Helwig, N. E. (2019). Statistical nonparametric mapping: Multivariate permutation tests for location, correlation, and regression problems in neuroimaging. *WIREs Computational Statistics*, 11(2), e1457. [10.1002/wics.1457](https://doi.org/10.1002/wics.1457)
- <http://users.stat.umn.edu/~helwig/notes/nptest-notes.html>

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# Need for Nonparametric Tests

Nonparametric randomization and permutation tests offer robust alternatives to classic (parametric) hypothesis tests.

Parametric hypothesis tests depend on parametric assumptions and/or large sample approximations for valid inference.

Nonparametric tests use computationally intensive methods to provide valid results under a wide collection of data generating conditions

Nonparametric tests are a form of *conditional inference*, where inference is conducted after conditioning on properties of the data.

# Reminder: Hypothesis Testing Basics

Null hypothesis significance testing procedure:

1. Form a null hypothesis  $H_0$  and an alternative hypothesis  $H_1$  about some parameter  $\theta = t(F)$  of a distribution  $F$
2. Calculate some test statistic  $T = s(\mathbf{x})$  from the observed data  $\mathbf{x} = (x_1, \dots, x_n)^\top$  where  $x_i \stackrel{\text{iid}}{\sim} F$  for all  $i = 1, \dots, n$
3. Calculate the p-value, which is the probability of observing a test statistic as or more extreme than  $T$  assuming that  $H_0$  is true
4. Reject  $H_0$  if the p-value is below some user-determined threshold, e.g.,  $\alpha = 0.05$  or some other small value

Notational notes:

- $\theta = t(F)$  emphasizes that parameters are functions of distributions
- $T = s(\mathbf{x})$  emphasizes that statistics are functions of sampled data

# Sampling Distribution and Null Sampling Distribution

The **sampling distribution** of  $T$  refers to the distribution of  $T$  that would be obtained by applying the test statistic function  $s(\cdot)$  to many different random samples of size  $n$  drawn from the population  $F$ .

The **null sampling distribution** of  $T$  is the sampling distribution under the assumption that  $H_0$  is true (used in step 3 to calculate p-value)

In parametric tests, the null sampling distribution is derived from

- parametric assumptions about the data ( $F$  is normal)
- large sample assumptions about the statistic ( $T$  is asymp normal)

# Nonparametric Tests use Randomization/Permutation

Nonparametric tests approximate the null sampling distribution by

1. enumerating all data permutations that are equally likely under  $H_0$
2. calculating the test statistic  $T$  for each possible data permutation

An **exact test** can be conducted by fully enumerating all possible data arrangements that are equally likely under  $H_0$

- One-Sample (or Paired Sample):  $M = 2^n$  possible arrangements
- Two-Sample (Independent):  $M = \binom{n_1+n_2}{n_1}$  possible arrangements

An **approximate test** can be conducted by randomly sampling  $R < M$  data arrangements from the collection of  $M$  possible arrangements.

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# Univariate Reminder: Student's One-Sample $t$ Test

Given  $x_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$  with  $\sigma^2$  unknown, suppose we want to test

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0$$

where  $\mu_0$  is some known value specified by the null hypothesis.

We use Student's  $t$  test, where the  $t$  test statistic is given by

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where  $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$  and  $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ .

Under  $H_0$ , the  $T$  test statistic follows a Student's  $t$  distribution with degrees of freedom  $\nu = n - 1$ .

# Multivariate Reminder: Hotelling's One-Sample $T$ Test

Given  $\mathbf{x}_i \stackrel{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  with  $\boldsymbol{\Sigma}$  is unknown, suppose we want to test

$$H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0 \quad \text{versus} \quad H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$$

where  $\boldsymbol{\mu}_0$  is some known vector specified by the null hypothesis.

Hotellings  $T^2$  is multivariate extension of (squared)  $t$  test statistic

$$T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)$$

where  $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$  and  $\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$

Under  $H_0$ , Hotelling's  $T^2$  follows a scaled  $F$  distribution

$$T^2 \sim \frac{(n-1)p}{(n-p)} F_{p, n-p}$$

and asymptotically follows a  $\chi_p^2$  distribution (even for non-normal data)

# Limitations of Hotelling's One-Sample $T$ Test

To calculate Hotelling's  $T^2$ , we need  $\mathbf{S}$  to be nonsingular

- Data matrix  $\mathbf{X} = [x_{ij}]$  needs to have linearly independent columns
- Implies that we need more observations than variables ( $n \geq p$ )

Exact inference requires multivariate normality assumption

- Multivariate normality is an idealization/approximation
- With real data, this assumption is wishful thinking

Asymptotic inference requires “large enough”  $n$  relative to  $p$

- How large does  $n$  need to be? It depends...
- May not be possible in high-dimensional settings

# Nonparametric One-Sample Location Problem

Given  $x_i \stackrel{\text{iid}}{\sim} F$  for some symmetric distribution  $F$  with mean  $\mu$  and variance  $\sigma^2$ , suppose we want to test

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0$$

where  $\mu_0$  is some known value specified by the null hypothesis.

Under  $H_0$ , the  $x_i$  are symmetric around  $\mu_0$ , which implies...

- the  $x_i - \mu_0$  are symmetric around 0
- $P(x_i > \mu_0) = P(x_i < \mu_0) = 1/2$

Note: procedure works asymptotically for asymmetric distributions too!

# Enumerating the Possible Data Configurations

Define  $d_i = x_i - \mu_0$  and consider writing the deviation scores as

$$d_i = \psi_i m_i$$

where

- $\psi_i = \text{sign}(x_i - \mu_0)$  is the sign of the deviation score
- $m_i = |x_i - \mu_0|$  is the magnitude of the deviation score

Given  $\{m_1, \dots, m_n\}$ , there are  $M = 2^n$  equally likely configurations:

- each  $\psi_i$  takes values  $\{-1, 1\}$  with equal probability
- the  $\psi_i$  are mutually independent of one another

# Permutations via Sign-Flipping Vectors

Let  $\boldsymbol{\psi}_k = (\psi_{1k}, \dots, \psi_{nk})^\top$  denote the  $k$ -th **sign-flipping vector**

- Note that  $\psi_{ik} \in \{-1, 1\}$  for all  $i = 1, \dots, n$
- $\{\boldsymbol{\psi}_1, \dots, \boldsymbol{\psi}_M\}$  denotes the collection of  $M = 2^n$  signing vectors

The  $k$ -th permuted (sign-flipped) version of the data is defined as

$$d_{ik} = \psi_{ik} d_i = \psi_{ik} \psi_i m_i$$

and the corresponding copy of the test statistic is defined as

$$T_k = \frac{\bar{d}_k}{s_{d_k} / \sqrt{n}}$$

where  $\bar{d}_k = \frac{1}{n} \sum_{i=1}^n d_{ik}$  and  $s_{d_k}^2 = \frac{1}{n-1} \sum_{i=1}^n (d_{ik} - \bar{d}_k)^2$

# Permutation Distribution and Inference

The set of test statistics  $\{T_k\}_{k=1}^M$  forms the **permutation distribution**, which is a reasonable surrogate for the null sampling distribution.

- Approximate test uses  $R < M$  random samples from  $\{T_k\}_{k=1}^M$

Calculate observed  $t$  test statistic

$$T_0 = \frac{\bar{d}}{s_d/\sqrt{n}}$$

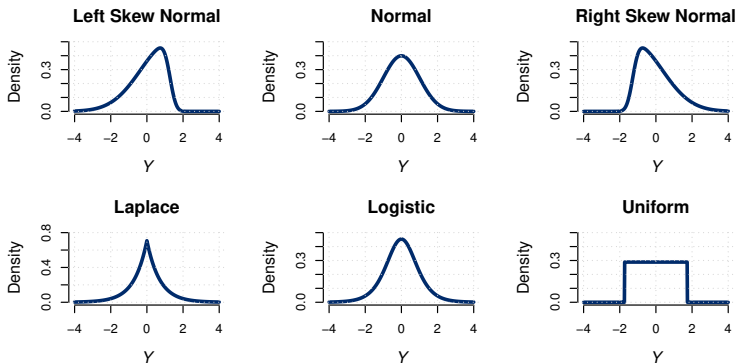
where  $\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i$  and  $s_d^2 = \frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2$ .

Calculate the p-value by comparing  $T_0$  to permutation distribution

- $H_1 : \mu \neq \mu_0$  (two-sided):  $p = \frac{1}{M} \sum_{k=1}^M I(|T_k| > |T_0|)$
- $H_1 : \mu < \mu_0$  (less than):  $p = \frac{1}{M} \sum_{k=1}^M I(T_k < T_0)$
- $H_1 : \mu > \mu_0$  (greater than):  $p = \frac{1}{M} \sum_{k=1}^M I(T_k > T_0)$

# Simulation Design from Helwig (2019)

Manipulated two factors: the distribution for (see Figure 1), and the sample size  $n \in \{10, 25, 50, 100, 200\}$ .



**Figure 1:** Six univariate distributions with mean  $\mu = 0$  and variance  $\sigma^2 = 1$ .



## Simulation Analyses from Helwig (2019)

10,000 independent datasets (replications) were generated for each of the 30 (6 distribution  $\times$  5  $n$ ) cells of the simulation design.

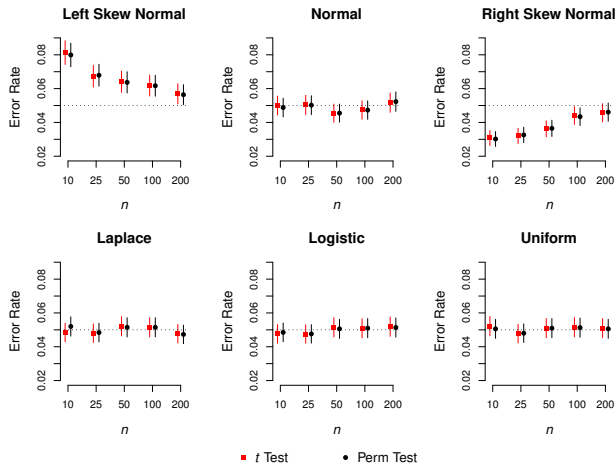
For each replication, the null hypothesis  $H_0 : \mu = 0$  was tested using the alternative hypothesis  $H_1 : \mu > 0$  and an  $\alpha = 0.05$  significance level.

The significance testing results were compared using two methods:

- (i) Student's  $t$  test
- (ii) Fisher's randomization test

The number of resamples was set at  $R = \min(2^n, 9999)$ .

# Simulation Results from Helwig (2019)



**Figure 2:** Type I error rate calculated across 10,000 replications for the  $t$  test (red squares) and the permutation test (black circles). The vertical bars denote approximate 99% confidence intervals, and the dotted horizontal lines denote the nominal rate of  $\alpha = 0.05$ .

# Extensions to Multivariate Tests

Given  $\mathbf{x}_i \stackrel{\text{iid}}{\sim} F$  for some symmetric distribution  $F$  with mean vector  $\boldsymbol{\mu}$  and covariance matrix  $\boldsymbol{\Sigma}$ , suppose we want to test

$$H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0 \quad \text{versus} \quad H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$$

where  $\boldsymbol{\mu}_0$  is some known vector specified by the null hypothesis.

To test  $H_0$ , consider a test statistic of the form

$$T_0^* = \max_j |T_j|$$

where  $T_j = \frac{\bar{x}_j - \mu_{j0}}{s_{jj}/\sqrt{n}}$  is Student's  $t$  test statistic for the  $j$ -th variable

- $\bar{x}_j$  and  $\mu_{j0}$  denote the  $j$ -th elements of  $\bar{\mathbf{x}}$  and  $\boldsymbol{\mu}_0$
- $s_{jj}$  denotes the  $j$ -th diagonal element of  $\mathbf{S}$

# Permutations via Multivariate Sign-Flipping

The  $k$ -th permuted (sign-flipped) version of the  $j$ -th variable is

$$d_{ijk} = \psi_{ik} d_{ij}$$

where  $\psi_{ik} \in \{-1, 1\}$  and  $d_{ij} = x_{ij} - \mu_{0j}$ .

Letting  $\bar{d}_{jk} = \frac{1}{n} \sum_{i=1}^n d_{ijk}$  and  $s_{d_{jk}}^2 = \frac{1}{n-1} \sum_{i=1}^n (d_{ijk} - \bar{d}_{jk})^2$ , define

$$T_{jk} = \frac{\bar{d}_{jk}}{s_{d_{jk}}/\sqrt{n}}$$

which is the  $j$ -th variable's (univariate) test statistic, and define

$$T_k^* = \max_j |T_{jk}|$$

which is the multivariate test statistic (for the  $k$ -th permutation).

# Permutation Distribution and Inference

The set of test statistics  $\{T_k^*\}_{k=1}^M$  forms the **permutation distribution**, which is a reasonable surrogate for the null sampling distribution.

- Approximate test uses  $R < M$  random samples from  $\{T_k^*\}_{k=1}^M$

Calculate the p-value by comparing  $T_0^*$  to permutation distribution

- $H_1 : \mu \neq \mu_0$  (two-sided):  $p = \frac{1}{M} \sum_{k=1}^M I(|T_k^*| > |T_0^*|)$
- $H_1 : \mu < \mu_0$  (less than):  $p = \frac{1}{M} \sum_{k=1}^M I(T_k^* < T_0^*)$
- $H_1 : \mu > \mu_0$  (greater than):  $p = \frac{1}{M} \sum_{k=1}^M I(T_k^* > T_0^*)$

For directional alternatives, the multivariate test statistic is defined as

- $H_1 : \mu < \mu_0$  (less than):  $T_0^* = \min_j T_j$  and  $T_k^* = \min_j T_{jk}$
- $H_1 : \mu > \mu_0$  (greater than):  $T_0^* = \max_j T_j$  and  $T_k^* = \max_j T_{jk}$

# Adjusted P-Values for Univariate Tests

The permutation distribution  $\{T_k^*\}_{k=1}^M$  can be used to compute **adjusted p-values** for testing the null hypotheses  $H_{0j} : \mu_j = \mu_{0j}$

Calculate the adjusted p-values by comparing  $T_j$  to  $\{T_k^*\}_{k=1}^M$

- $H_1 : \mu_j \neq \mu_{0j}$  (two-sided):  $p = \frac{1}{M} \sum_{k=1}^M I(|T_k^*| > |T_j|)$
- $H_1 : \mu_j < \mu_{0j}$  (less than):  $p = \frac{1}{M} \sum_{k=1}^M I(T_k^* < T_j)$
- $H_1 : \mu_j > \mu_{0j}$  (greater than):  $p = \frac{1}{M} \sum_{k=1}^M I(T_k^* > T_j)$

Adjusted p-values control the familywise error rate across the  $p$  tests

- Not exact control (i.e., FWER will not exceed  $\alpha$  level)

## `np.loc.test()` Function in **npreg** R Package

The `np.loc.test()` function in the **npreg** R package is designed to work similarly to the `t.test()` function in R's **stats** package:

```
np.loc.test(x, y = NULL,  
            alternative = c("two.sided", "less", "greater"),  
            mu = 0, paired = FALSE, var.equal = FALSE,  
            median.test = FALSE, symmetric = TRUE,  
            R = 9999, parallel = FALSE, cl = NULL,  
            perm.dist = TRUE, na.rm = TRUE)
```

But it offers a lot more options, e.g., median tests, test statistics for asymmetric distributions, and multivariate tests.

## Example: $H_0$ True and Small Sample Size ( $n = 10$ )

```
> library(nptest)
> n <- 10
> p <- 4
> set.seed(1)
> X <- replicate(p, rlogis(n))      # logistic distribution
> test <- np.loc.test(X)
> test
```

Multivariate Nonparametric Location Test (One Sample t-test)

alternative hypothesis: true mean is not equal to 0

t = 0.8929, p-value = 0.8945

sample estimates:

| mu1        | mu2        | mu3         | mu4        |
|------------|------------|-------------|------------|
| 0.31610391 | 0.51257065 | -0.49803032 | 0.09427151 |

```
> test$adj.p.values
[1] 0.9863281 0.8945312 0.9238281 1.0000000
```



## Example: $H_0$ True and Moderate Sample Size ( $n = 100$ )

```
> library(nptest)
> n <- 100
> p <- 4
> set.seed(1)
> X <- replicate(p, rlogis(n))      # logistic distribution
> test <- np.loc.test(X)
> test
```

Multivariate Nonparametric Location Test (One Sample t-test)

alternative hypothesis: true mean is not equal to 0

t = 1.783, p-value = 0.2737

sample estimates:

| mu1        | mu2        | mu3         | mu4        |
|------------|------------|-------------|------------|
| 0.08733611 | 0.14315329 | -0.28262220 | 0.17138996 |

```
> test$adj.p.values
[1] 0.9656 0.8493 0.2737 0.8084
```

## Example: $H_0$ False and Small Sample Size ( $n = 10$ )

```
> n <- 10
> p <- 4
> set.seed(1)
> mu <- c(0,0,1,1)
> X <- replicate(p, rlogis(n)) + matrix(mu, n, p, byrow = TRUE)
> test <- np.loc.test(X)
> test
```

Multivariate Nonparametric Location Test (One Sample t-test)

alternative hypothesis: true mean is not equal to 0

t = 2.9312, p-value = 0.0625

sample estimates:

| mu1       | mu2       | mu3       | mu4       |
|-----------|-----------|-----------|-----------|
| 0.3161039 | 0.5125707 | 0.5019697 | 1.0942715 |

```
> test$adj.p.values
[1] 0.9804688 0.8945312 0.9316406 0.0625000
```

## Example: $H_0$ False and Moderate Sample Size ( $n = 100$ )

```
> n <- 100
> p <- 4
> set.seed(1)
> mu <- c(0,0,1,1)
> X <- replicate(p, rlogis(n)) + matrix(mu, n, p, byrow = TRUE)
> test <- np.loc.test(X)
> test
```

Multivariate Nonparametric Location Test (One Sample t-test)

alternative hypothesis: true mean is not equal to 0

t = 6.5325, p-value = 1e-04

sample estimates:

|  | mu1        | mu2        | mu3        | mu4        |
|--|------------|------------|------------|------------|
|  | 0.08733611 | 0.14315329 | 0.71737780 | 1.17138996 |

```
> test$adj.p.values
[1] 0.9632 0.8442 0.0001 0.0001
```

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# Univariate Reminder: Dependent Samples $t$ Test

Given  $d_i \stackrel{\text{iid}}{\sim} N(\mu_d, \sigma_d^2)$  with  $\sigma_d^2$  unknown, suppose we want to test

$$H_0 : \mu_d = \mu_0 \quad \text{versus} \quad H_1 : \mu_d \neq \mu_0$$

where  $\mu_0$  is some known value specified by the null hypothesis. Note:

- $d_i = x_{1i} - x_{2i}$  and  $\mu_d = \mu_1 - \mu_2$  and  $\sigma_d^2 = \sigma_{11} + \sigma_{22} - 2\sigma_{12}$

We use Student's  $t$  test, where the  $t$  test statistic is given by

$$T = \frac{\bar{d} - \mu_0}{s_d / \sqrt{n}}$$

where  $\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i$  and  $s_d^2 = \frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2$  with  $d_i = x_{1i} - x_{2i}$ .

Under  $H_0$ , the  $T$  test statistic follows a Student's  $t$  distribution with degrees of freedom  $\nu = n - 1$ .

# Multivariate Reminder: Dependent Samples $T$ Test

Given  $\mathbf{d}_i \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}_d, \boldsymbol{\Sigma}_d)$  with  $\boldsymbol{\Sigma}_d$  unknown, suppose we want to test

$$H_0 : \boldsymbol{\mu}_d = \boldsymbol{\mu}_0 \quad \text{versus} \quad H_1 : \boldsymbol{\mu}_d \neq \boldsymbol{\mu}_0$$

where  $\boldsymbol{\mu}_0$  is some known value specified by the null hypothesis. Note:

- $\mathbf{d}_i = \mathbf{x}_{1i} - \mathbf{x}_{2i}$  and  $\boldsymbol{\mu}_d = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$  and  $\boldsymbol{\Sigma}_d = \boldsymbol{\Sigma}_{11} + \boldsymbol{\Sigma}_{22} - \boldsymbol{\Sigma}_{12} - \boldsymbol{\Sigma}_{21}$

We use Hotelling's  $T^2$  test, where the  $T^2$  test statistic is given by

$$T_d^2 = n(\bar{\mathbf{d}} - \boldsymbol{\mu}_d)^\top \mathbf{S}_d^{-1}(\bar{\mathbf{d}} - \boldsymbol{\mu}_d)$$

where  $\bar{\mathbf{d}} = \frac{1}{n} \sum_{i=1}^n \mathbf{d}_i$  and  $\mathbf{S}_d = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{d}_i - \bar{\mathbf{d}})(\mathbf{d}_i - \bar{\mathbf{d}})^\top$

If  $H_0$  is true,  $T^2$  follows a scaled  $F$  distribution:  $T^2 \sim \frac{(n-1)p}{n-p} F_{p, n-p}$

# Nonparametric Multivariate Dependent Samples $T$ Test

Same as one sample test on a vector of difference scores.

- $\mathbf{d}_i = \mathbf{x}_{1i} - \mathbf{x}_{2i}$  is the difference score vector
- Apply NPMV one sample test with  $\mathbf{d}_i$  in place of  $\mathbf{x}_i$

Two syntactical options for conducting the test using `np.loc.test()`

- Use `x` and `y` inputs with `paired = TRUE` (preferred approach)
- Use `x` argument only by directly inputting the difference scores

Note: both code options will return the same result, but some text in the R console printout will differ depending on the type.

## Example: Preferred Syntax

```
> n <- 100
> p <- 4
> set.seed(1)
> X <- replicate(p, rlogis(n))
> Y <- replicate(p, rlogis(n))
> test <- np.loc.test(X, Y, paired = TRUE)
> test
```

Multivariate Nonparametric Location Test (Paired t-test)

alternative hypothesis: true difference of means is not equal to 0

t = 1.8389, p-value = 0.2501

sample estimates:

| mu1        | mu2         | mu3         | mu4        |
|------------|-------------|-------------|------------|
| 0.09837344 | -0.03758638 | -0.40142985 | 0.19200488 |

```
> test$adj.p.values
[1] 0.9910 0.9998 0.2501 0.9334
```



## Example: Alternative Syntax

```
> n <- 100
> p <- 4
> set.seed(1)
> X <- replicate(p, rlogis(n))
> Y <- replicate(p, rlogis(n))
> test <- np.loc.test(X - Y)
> test
```

Multivariate Nonparametric Location Test (One Sample t-test)

alternative hypothesis: true mean is not equal to 0

t = 1.8389, p-value = 0.2501

sample estimates:

| mu1        | mu2         | mu3         | mu4        |
|------------|-------------|-------------|------------|
| 0.09837344 | -0.03758638 | -0.40142985 | 0.19200488 |

```
> test$adj.p.values
[1] 0.9910 0.9998 0.2501 0.9334
```

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# Univariate Reminder: Independent Samples $t$ Test

Given  $x_{1i} \stackrel{\text{iid}}{\sim} N(\mu_1, \sigma^2)$  and  $x_{2i} \stackrel{\text{iid}}{\sim} N(\mu_2, \sigma^2)$  with  $\sigma^2$  unknown

- Hypotheses:  $H_0 : \mu_1 - \mu_2 = \mu_0$  versus  $H_1 : \mu_1 - \mu_2 \neq \mu_0$
- Test statistic:  $T = \frac{\bar{x}_1 - \bar{x}_2 - \mu_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$  where  $\bar{x}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} x_{ik}$ ,  
 $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}$ , and  $s_k^2 = \frac{1}{n_k-1} \sum_{i=1}^{n_k} (x_{ik} - \bar{x}_k)^2$
- Inference: When  $H_0$  is true,  $T \sim t_\nu$  with  $\nu = n_1 + n_2 - 2$

Given  $x_{1i} \stackrel{\text{iid}}{\sim} N(\mu_1, \sigma_1^2)$  and  $x_{2i} \stackrel{\text{iid}}{\sim} N(\mu_2, \sigma_2^2)$  with  $\sigma_k^2$  unknown

- Hypotheses:  $H_0 : \mu_1 - \mu_2 = \mu_0$  versus  $H_1 : \mu_1 - \mu_2 \neq \mu_0$
- Test statistic:  $T = \frac{\bar{x}_1 - \bar{x}_2 - \mu_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \approx t_\nu$  with  $\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1-1} \left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2-1} \left(\frac{s_2^2}{n_2}\right)^2}$
- Inference: When  $H_0$  is true,  $T$  approximately follows Student's  $t_\nu$

# Multivariate Reminder: Independent Samples $T$ Test

Given  $\mathbf{x}_{ki} \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$  for  $k \in \{1, 2\}$  with  $\boldsymbol{\Sigma}$  unknown

- Hypotheses:  $H_0 : \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 = \boldsymbol{\mu}_0$  versus  $H_1 : \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 \neq \boldsymbol{\mu}_0$
- Test statistic:  $T^2 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - \boldsymbol{\mu}_0)^\top \left[ \left( \frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_p \right]^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - \boldsymbol{\mu}_0)$   
 $\mathbf{S}_p = \frac{n_1-1}{n_1+n_2-2} \mathbf{S}_1 + \frac{n_2-1}{n_1+n_2-2} \mathbf{S}_2$  and  $\mathbf{S}_k = \frac{1}{n_k-1} \sum_{i=1}^{n_k} (\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)(\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)^\top$
- Inference: When  $H_0$  is true,  $T^2 \sim \frac{(n_1+n_2-2)p}{n_1+n_2-p-1} F_{p, n_1+n_2-p-1}$

Given  $\mathbf{x}_{ki} \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$  for  $k \in \{1, 2\}$  with  $\boldsymbol{\Sigma}_k$  unknown

- Hypotheses:  $H_0 : \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 = \boldsymbol{\mu}_0$  versus  $H_1 : \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 \neq \boldsymbol{\mu}_0$
- Test statistic:  $T^2 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - \boldsymbol{\mu}_0)^\top \mathbf{S}_0^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - \boldsymbol{\mu}_0)$  where the non-pooled covariance matrix is  $\mathbf{S}_0 = n_1^{-1} \mathbf{S}_1 + n_2^{-1} \mathbf{S}_2$
- Inference: When  $H_0$  is true,  $T^2 \approx \frac{\nu p}{\nu - p + 1} F_{p, \nu - p + 1}$  where  

$$\nu = (p + p^2) \left( \sum_{k=1}^2 \frac{1}{n_k} \left\{ \text{tr} \left[ \left( \frac{1}{n_k} \mathbf{S}_k \mathbf{S}_0^{-1} \right)^2 \right] + \left( \text{tr} \left[ \frac{1}{n_k} \mathbf{S}_k \mathbf{S}_0^{-1} \right] \right)^2 \right\} \right)^{-1}$$

# Nonparametric Two-Sample Location Problem

Given  $x_{1i} \stackrel{\text{iid}}{\sim} F_1$  and  $x_{2i} \stackrel{\text{iid}}{\sim} F_2$  where  $F_k(z) = F_0(z - \mu_k)$  for  $k \in \{1, 2\}$ , suppose we want to test

$$H_0 : \mu_1 - \mu_2 = \mu_0 \quad \text{versus} \quad H_1 : \mu_1 - \mu_2 \neq \mu_0$$

which is equivalent to testing  $H_0 : F_1 = F_2$  if the null value is  $\mu_0 = 0$ .

We could consider the more general (location-scale) model

$$F_k(z) = F_0\left(\frac{z - \mu_k}{\sigma_k}\right)$$

where  $\mu_k$  are the location parameters, and  $\sigma_k$  are the scale parameters for  $k \in \{1, 2\}$ . In this case,  $F_1 = F_2$  when  $\mu_1 = \mu_2$  and  $\sigma_1 = \sigma_2$ .

# Enumerating the Possible Data Configurations

Let  $\{x_{1i}\}_{i=1}^{n_1}$  and  $\{x_{2i}\}_{i=1}^{n_2}$  denote independent samples

- $\{(z_i, g_i)\}_{i=1}^N$  denotes the observed data from both groups
- $\{z_i\}_{i=1}^N$  is the combined sample where  $N = n_1 + n_2$
- $\{g_i\}_{i=1}^N$  denotes the group membership ( $g_i \in \{1, 2\}$ )

Given  $\{z_1, \dots, z_N\}$ , there are  $M = \binom{n_1+n_2}{n_1}$  equally likely configurations

- each  $g_i$  takes values  $\{1, 2\}$  which are arbitrary labels under  $H_0$
- exact test requires assuming the local-scale model with  $\sigma_1 = \sigma_2$
- asymptotically works when  $\sigma_1 \neq \sigma_2$  (if correct test statistic is used)

# Permutations via Group-Shuffling Vectors

Let  $\mathbf{g}_k = (g_{1k}, \dots, g_{Nk})^\top$  denote the  $k$ -th **group-shuffling** vector

- Note that  $g_{ik} \in \{1, 2\}$  for all  $i = 1, \dots, n$
- $\{\mathbf{g}_1, \dots, \mathbf{g}_M\}$  denotes the collection of  $M = \binom{N}{n_1}$  grouping vectors

The  $k$ -th permuted (group-shuffled) version of the data is defined as  $\{(z_i, g_{ik})\}_{i=1}^N$ , and the corresponding copy of the test statistic is

$$T_k = \frac{\bar{z}_{1k} - \bar{z}_{2k} - \mu_0}{s_{0k} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{or} \quad T_k = \frac{\bar{z}_{1k} - \bar{z}_{2k} - \mu_0}{\sqrt{\frac{s_{1k}^2}{n_1} + \frac{s_{2k}^2}{n_2}}}$$

where

- $\bar{z}_{1k} = \frac{1}{n_1} \sum_{i=1}^N z_i I(g_{ik} = 1)$  and  $\bar{z}_{2k} = \frac{1}{n_2} \sum_{i=1}^N z_i I(g_{ik} = 2)$
- $s_{0k}^2 = [(n_1 - 1)s_{1k}^2 + (n_2 - 1)s_{2k}^2] / (N - 2)$  is pooled variance
- $s_{1k}^2 = \frac{1}{n_1 - 1} \sum_{i=1}^N I(g_{ik} = 1)(z_i - \bar{z}_{1k})^2$  and  $s_{2k}^2 = \frac{1}{n_2 - 1} \sum_{i=1}^N I(g_{ik} = 2)(z_i - \bar{z}_{2k})^2$

# Permutation Distribution and Inference

The set of test statistics  $\{T_k\}_{k=1}^M$  forms the **permutation distribution**, which is a reasonable surrogate for the null sampling distribution.

- Approximate test uses  $R < M$  random samples from  $\{T_k\}_{k=1}^M$

Calculate observed  $t$  test statistic

$$T_0 = \frac{\bar{x}_1 - \bar{x}_2 - \mu_0}{s_0 \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{or} \quad T_0 = \frac{\bar{x}_1 - \bar{x}_2 - \mu_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where  $\bar{x}_j = \frac{1}{n_j} \sum_{i=1}^{n_j} x_{ji}$  and  $s_j^2 = \frac{1}{n_j-1} \sum_{i=1}^{n_j} (x_{ji} - \bar{x}_j)^2$  for  $j \in \{1, 2\}$ , and  $s_0^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$  is the pooled variance estimate.

Calculate the p-value by comparing  $T_0$  to permutation distribution

- $H_1 : \mu_1 - \mu_2 \neq \mu_0$  (two-sided):  $p = \frac{1}{M} \sum_{k=1}^M I(|T_k| > |T_0|)$
- $H_1 : \mu_1 - \mu_2 < \mu_0$  (less than):  $p = \frac{1}{M} \sum_{k=1}^M I(T_k < T_0)$
- $H_1 : \mu_1 - \mu_2 > \mu_0$  (greater than):  $p = \frac{1}{M} \sum_{k=1}^M I(T_k > T_0)$



# Simulation Design from Helwig (2019)

Manipulated three factors:

- (i) the data generating distribution (see top row of Figure 1)
- (ii) the standard deviation of the second group:  $\sigma_2 \in \{0.5, 1, 2\}$
- (iii) the sample size of the first group:  $n_1 \in \{10, 25, 50, 100, 200\}$

Throughout the simulation...

- (a) the standard deviation of the first group was fixed at  $\sigma_1 = 1$
- (b) the sample size of the second group was defined as  $n_2 = 2n_1$

## Simulation Analyses from Helwig (2019)

10,000 independent datasets (replications) were generated for each of the 45 ( $3 \text{ distribution} \times 3 \sigma_2 \times 5 n_1$ ) cells of the simulation design.

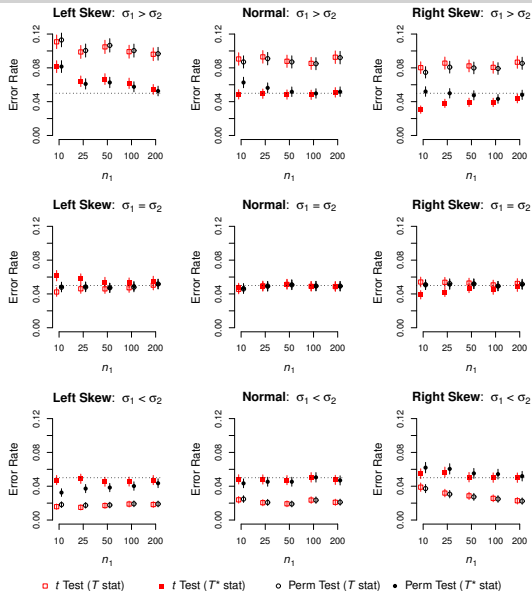
For each replication, the null hypothesis  $H_0 : \mu_1 = \mu_2$  was tested using the alternative hypothesis  $H_1 : \mu_1 > \mu_2$  and an  $\alpha = 0.05$  level.

The significance testing results were compared using four methods:

- (i) Student's  $t$  test which assumes normality and equal variances
- (ii) Welch's  $t$  test which assumes normality and unequal variances
- (iii) permutation test using Student's (pooled variance)  $T$  test statistic
- (iv) permutation test using Welch's (unequal variance)  $T^*$  test statistic

The number of resamples was set at  $R = 9999$ .

# Simulation Results from Helwig (2019)



**Figure 3:** Type I error rate calculated across 10,000 replications for the  $t$  test (red squares) and the permutation test (black circles). The unfilled points denote results using the (Student)  $T$  test statistic, whereas the filled points denote results using the (Welch)  $T^*$  test statistic. The vertical bars denote approximate 99% confidence intervals, and the dotted horizontal lines denote the nominal rate of  $\alpha = 0.05$ .

# Extensions to Multivariate Tests

Given  $\mathbf{x}_{1i} \stackrel{\text{iid}}{\sim} F_1$  and  $\mathbf{x}_{2i} \stackrel{\text{iid}}{\sim} F_2$  where  $F_k(\mathbf{z}) = F_0(\mathbf{z} - \boldsymbol{\mu}_k)$  for  $k \in \{1, 2\}$ , suppose we want to test

$$H_0 : \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 = \boldsymbol{\mu}_0 \quad \text{versus} \quad H_1 : \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 \neq \boldsymbol{\mu}_0$$

which is equivalent to testing  $H_0 : F_1 = F_2$  if the null value is  $\boldsymbol{\mu}_0 = \mathbf{0}_p$ .

To test  $H_0$ , consider a test statistic of the form

$$T_0^* = \max_j |T_j|$$

where  $T_j$  is the  $t$  test statistic for the  $j$ -th variable for  $j = 1, \dots, p$

$$\begin{aligned} \bullet \quad T_j &= \frac{\bar{x}_{1j} - \bar{x}_{2j} - \mu_{0j}}{s_{0j} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} & \text{where} \quad s_{0j}^2 &= \frac{(n_1 - 1)s_{1j}^2 + (n_2 - 1)s_{2j}^2}{n_1 + n_2 - 2} \\ \bullet \quad T_j &= \frac{\bar{x}_{1j} - \bar{x}_{2j} - \mu_{0j}}{\sqrt{\frac{s_{1j}^2}{n_1} + \frac{s_{2j}^2}{n_2}}} & \text{when} \quad \sigma_1 &\neq \sigma_2 \end{aligned}$$

# Permutations via Multivariate Group Shuffling

The  $k$ -th permuted (group-shuffled) version of the data is

$$\{(\mathbf{z}_i, g_{ik})\}_{i=1}^N$$

where  $\mathbf{z}_i$  is a row of the combined  $(N \times p)$  data matrix, and  $g_{ik}$  is the  $i$ -th element of the  $k$ -th group-shuffling vector  $\mathbf{g}_k$

The  $j$ -th variable's (univariate) test statistic for the  $k$ -th permutation is

$$T_{jk} = \frac{\bar{z}_{1jk} - \bar{z}_{2jk} - \mu_{0j}}{s_{0jk} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{or} \quad T_{jk} = \frac{\bar{z}_{1jk} - \bar{z}_{2jk} - \mu_{0j}}{\sqrt{\frac{s_{1jk}^2}{n_1} + \frac{s_{2jk}^2}{n_2}}}$$

and the multivariate test statistic for the  $k$ -th permutation is

$$T_k^* = \max_j |T_{jk}|$$

# Permutation Distribution and Inference

The set of test statistics  $\{T_k^*\}_{k=1}^M$  forms the **permutation distribution**, which is a reasonable surrogate for the null sampling distribution.

- Approximate test uses  $R < M$  random samples from  $\{T_k^*\}_{k=1}^M$

Calculate the p-value by comparing  $T_0^*$  to permutation distribution

- $H_1 : \mu_1 - \mu_2 \neq \mu_0$  (two-sided):  $p = \frac{1}{M} \sum_{k=1}^M I(|T_k^*| > |T_0^*|)$
- $H_1 : \mu_1 - \mu_2 < \mu_0$  (less than):  $p = \frac{1}{M} \sum_{k=1}^M I(T_k^* < T_0^*)$
- $H_1 : \mu_1 - \mu_2 > \mu_0$  (greater than):  $p = \frac{1}{M} \sum_{k=1}^M I(T_k^* > T_0^*)$

For directional alternatives, the multivariate test statistic is defined as

- $H_1 : \mu_{1j} - \mu_{2j} < \mu_{0j} \ \forall j$  (less than):  $T_0^* = \min_j T_j$  and  $T_k^* = \min_j T_{jk}$
- $H_1 : \mu_{1j} - \mu_{2j} > \mu_{0j} \ \forall j$  (greater than):  $T_0^* = \max_j T_j$  and  $T_k^* = \max_j T_{jk}$

# Adjusted P-Values for Univariate Tests

The permutation distribution  $\{T_k^*\}_{k=1}^M$  can be used to compute **adjusted p-values** for testing the null hypotheses  $H_{0j} : \mu_{1j} - \mu_{2j} = \mu_{0j}$

Calculate the adjusted p-values by comparing  $T_j$  to  $\{T_k^*\}_{k=1}^M$

- $H_1 : \mu_{1j} - \mu_{2j} \neq \mu_{0j}$  (two-sided):  $p = \frac{1}{M} \sum_{k=1}^M I(|T_k^*| > |T_j|)$
- $H_1 : \mu_{1j} - \mu_{2j} < \mu_{0j}$  (less than):  $p = \frac{1}{M} \sum_{k=1}^M I(T_k^* < T_j)$
- $H_1 : \mu_{1j} - \mu_{2j} > \mu_{0j}$  (greater than):  $p = \frac{1}{M} \sum_{k=1}^M I(T_k^* > T_j)$

Adjusted p-values control the familywise error rate across the  $p$  tests

- Not exact control (i.e., FWER will not exceed  $\alpha$  level)

## Example: $H_0$ True and Small Sample Size

```
> n <- c(10,12)
> p <- 4
> set.seed(1)
> X <- replicate(p, rlogis(n[1]))
> Y <- replicate(p, rlogis(n[2]))
> test <- np.loc.test(X, Y)
> test
```

Multivariate Nonparametric Location Test (Welch Two Sample t-test)  
 alternative hypothesis: true difference of means is not equal to 0  
 t = 1.4784, p-value = 0.5031  
 sample estimates:

| mu1         | mu2        | mu3         | mu4         |
|-------------|------------|-------------|-------------|
| -0.08342723 | 1.00809634 | -0.65733921 | -0.24326089 |

```
> test$adj.p.values
[1] 0.9999 0.5031 0.8658 0.9885
```



## Example: $H_0$ True and Moderate Sample Size

```
> n <- c(100,120)
> p <- 4
> set.seed(1)
> X <- replicate(p, rlogis(n[1]))
> Y <- replicate(p, rlogis(n[2]))
> test <- np.loc.test(X, Y)
> test
```

Multivariate Nonparametric Location Test (Welch Two Sample t-test)  
 alternative hypothesis: true difference of means is not equal to 0  
 t = 1.1509, p-value = 0.6858  
 sample estimates:

| mu1       | mu2        | mu3        | mu4       |
|-----------|------------|------------|-----------|
| 0.0432874 | -0.0132806 | -0.2488341 | 0.1035578 |

```
> test$adj.p.values
[1] 0.9995 1.0000 0.6858 0.9904
```

## Example: $H_0$ False and Small Sample Size

```
> n <- c(10,12)
> p <- 4
> mu <- c(0,0,1,1)
> set.seed(1)
> X <- replicate(p, rlogis(n[1])) + matrix(mu, n[1], p, byrow = TRUE)
> Y <- replicate(p, rlogis(n[2]))
> test <- np.loc.test(X, Y)
> test
```

Multivariate Nonparametric Location Test (Welch Two Sample t-test)  
 alternative hypothesis: true difference of means is not equal to 0  
 t = 1.4784, p-value = 0.5076  
 sample estimates:

|  | mu1         | mu2        | mu3        | mu4        |
|--|-------------|------------|------------|------------|
|  | -0.08342723 | 1.00809634 | 0.34266079 | 0.75673911 |

```
> test$adj.p.values
[1] 1.0000 0.5076 0.9865 0.6002
```

## Example: $H_0$ False and Moderate Sample Size

```
> n <- c(100,120)
> p <- 4
> mu <- c(0,0,1,1)
> set.seed(1)
> X <- replicate(p, rlogis(n[1])) + matrix(mu, n[1], p, byrow = TRUE)
> Y <- replicate(p, rlogis(n[2]))
> test <- np.loc.test(X, Y)
> test
```

Multivariate Nonparametric Location Test (Welch Two Sample t-test)  
 alternative hypothesis: true difference of means is not equal to 0  
 t = 4.2758, p-value = 1e-04  
 sample estimates:

| mu1       | mu2        | mu3       | mu4       |
|-----------|------------|-----------|-----------|
| 0.0432874 | -0.0132806 | 0.7511659 | 1.1035578 |

```
> test$adj.p.values
[1] 0.9993 1.0000 0.0023 0.0001
```